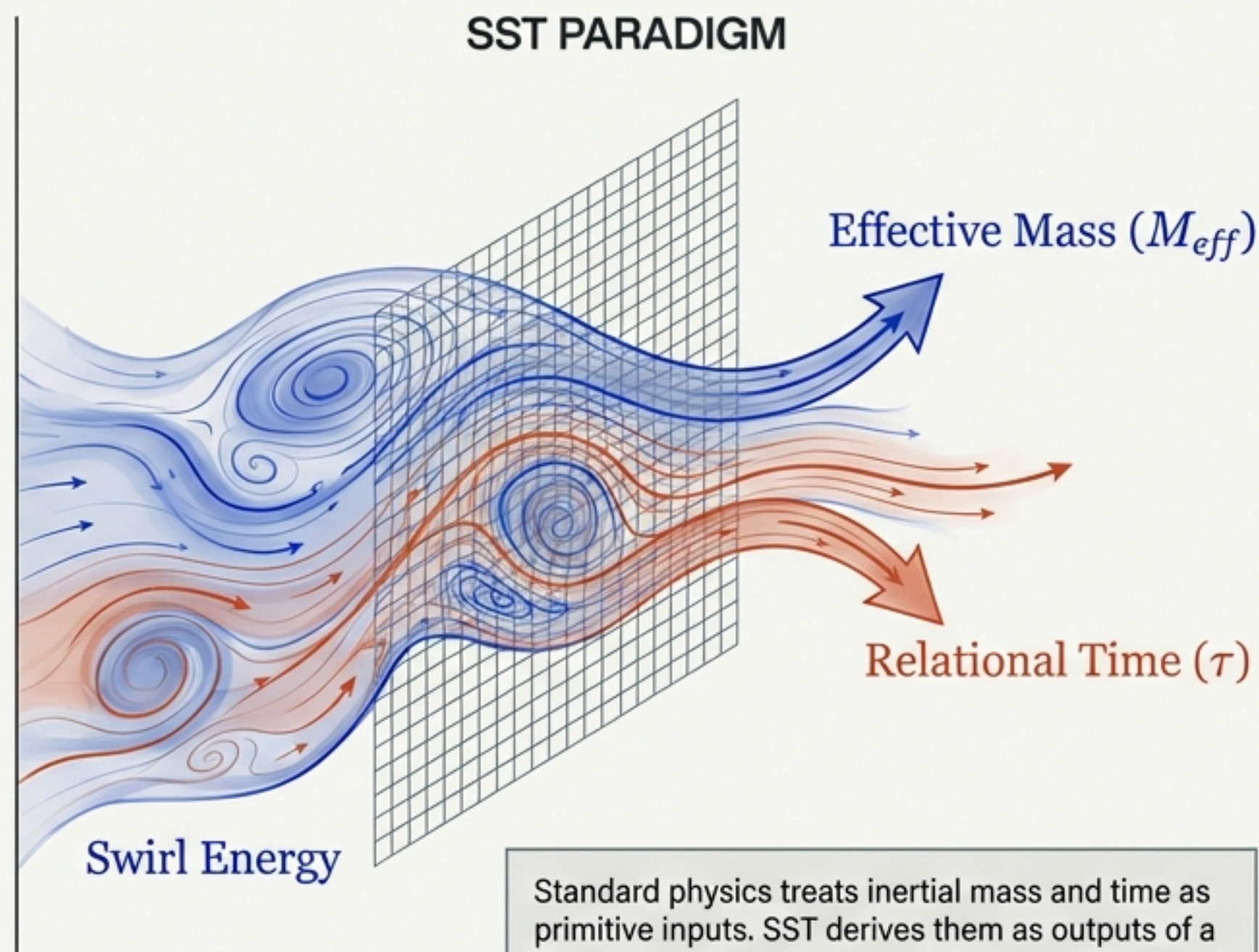
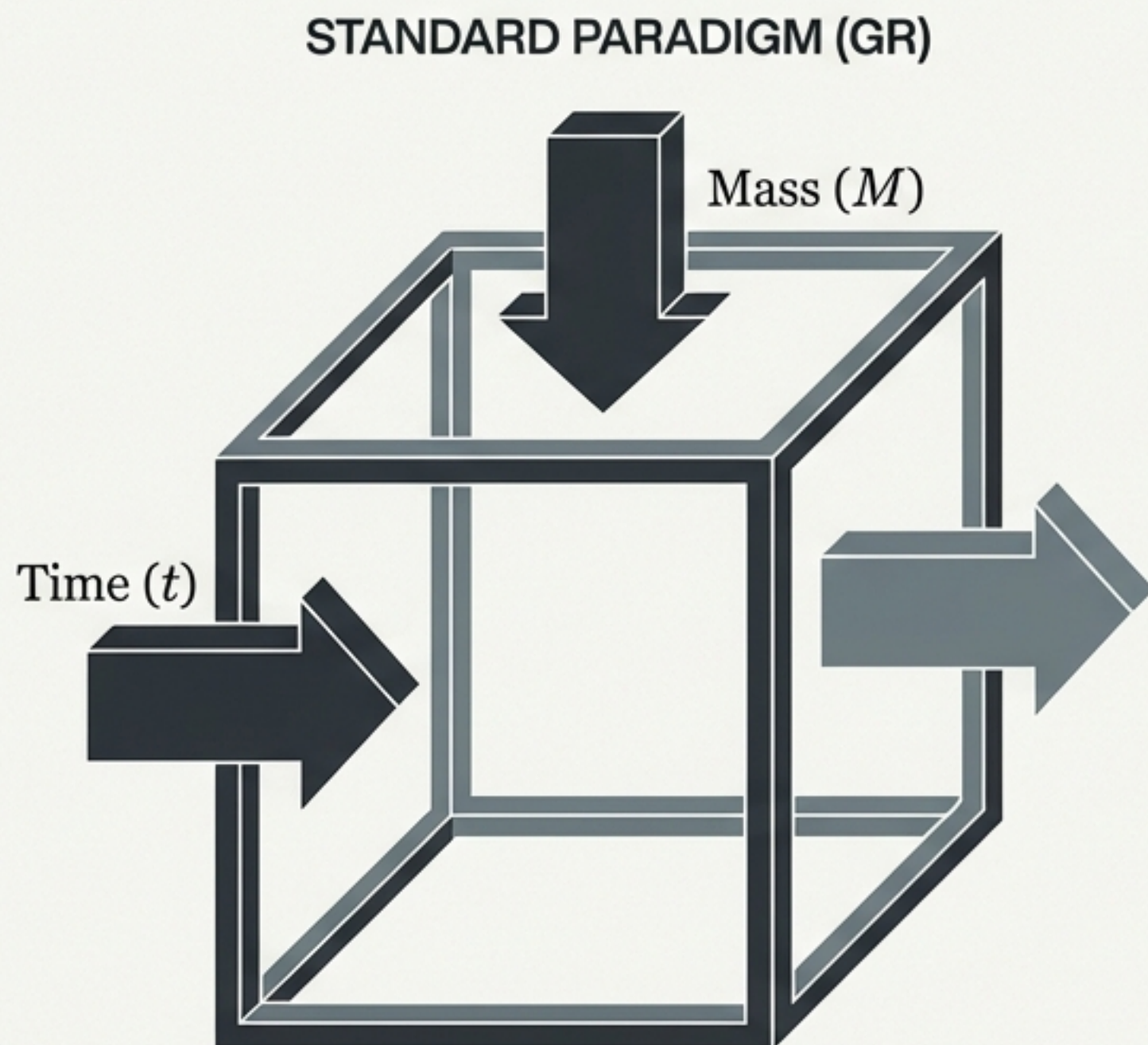


Velocity-Dependent Mass and the Texture of Time

Operational Consequences of Swirl-String Theory (SST-65)



Standard physics treats inertial mass and time as primitive inputs. SST derives them as outputs of a 'Swirl-String' medium and a 'Clock-Foliation.' This shift creates observable, testable anomalies in the weak-field limit.

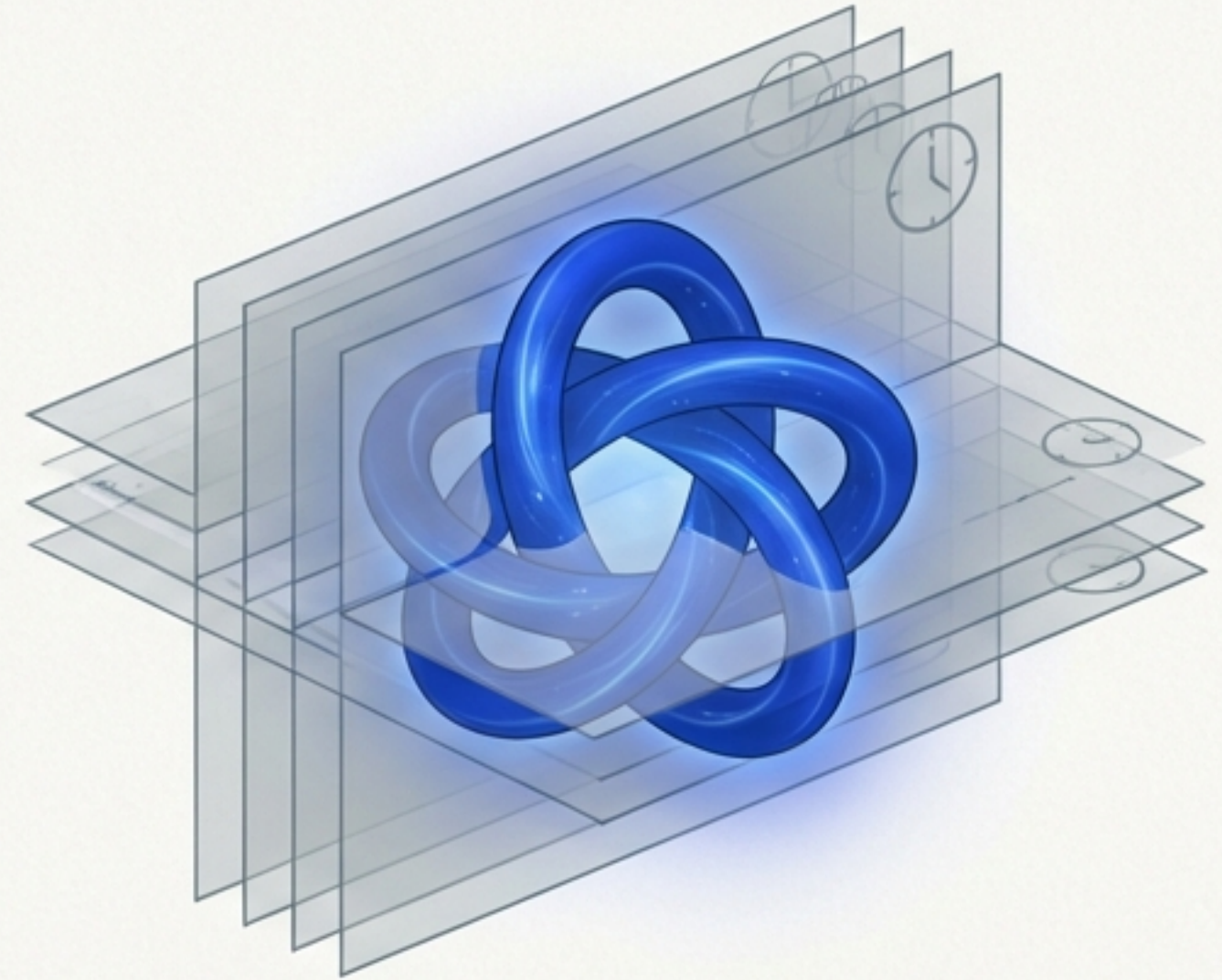
The Master Mass Functional: Deriving Instead of Assuming

In SST, inertial and gravitational mass are not intrinsic properties. They are defined by the integrated swirl energy density referenced to a local clock-foliation structure.

$$M = C \int_V \rho_E(x) dV$$

Annotations for the equation:

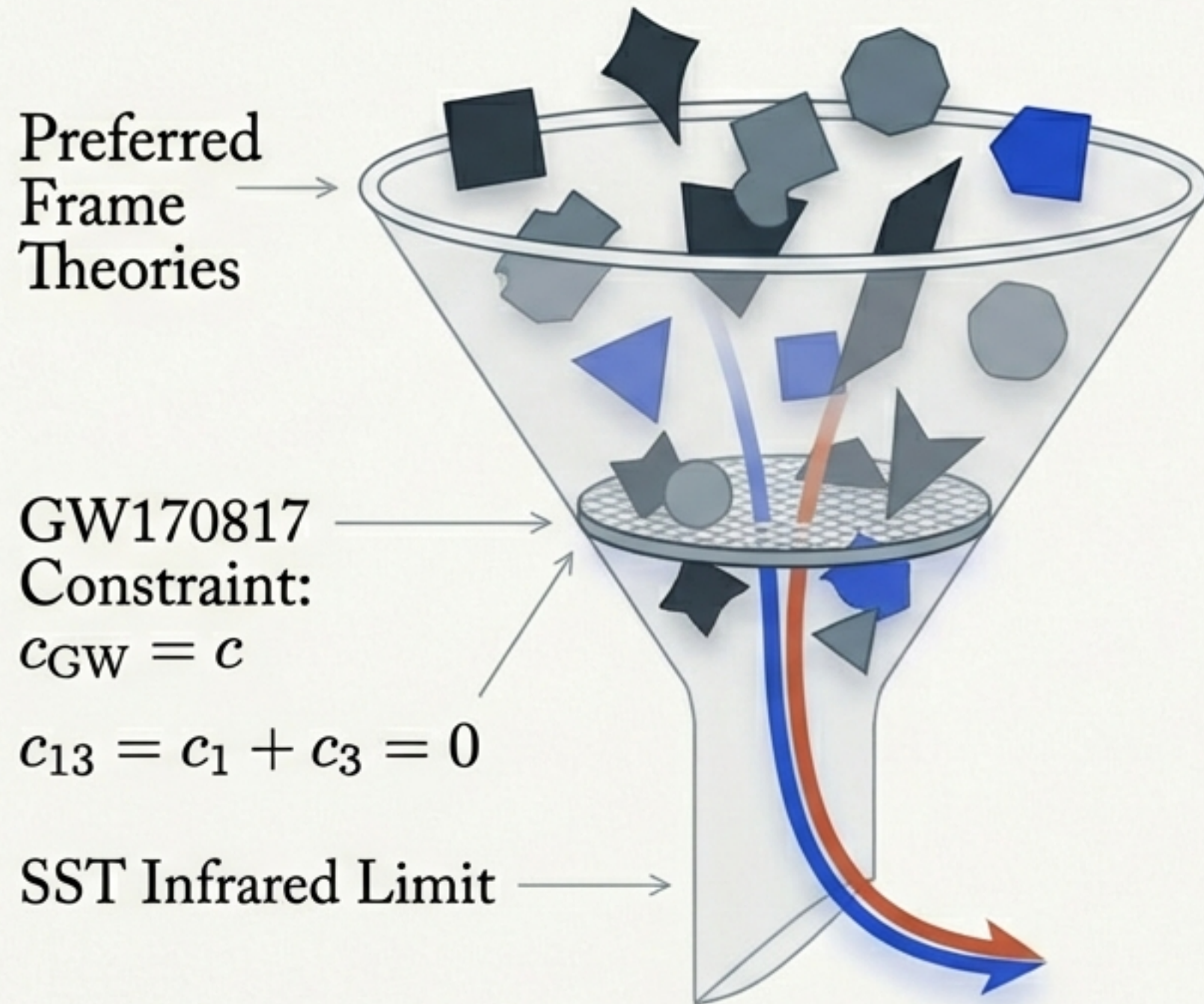
- Normalization Constant (points to C)
- Local Swirl Energy Density (points to $\rho_E(x)$)
- Integration relative to background clock field (points to dV)



Implication: Because mass is an integral over a background medium, it is susceptible to “slicing” effects. Moving through the medium changes the angle of integration, effectively altering the mass.

The Clock Foliation and Gravitational Constraints

Passing the GW170817 Filter



The background structure is a unit timelike vector field u^μ . To survive, SST must obey the observation that gravitational waves propagate at the speed of light.

Result: The preferred-frame parameters reduce to a narrow sector (c_{14} and c_2). SST is consistent with current data but predicts a residual anisotropic effects in the mass sector.

Operational Consequence: The Velocity-Dependent Mass Functional

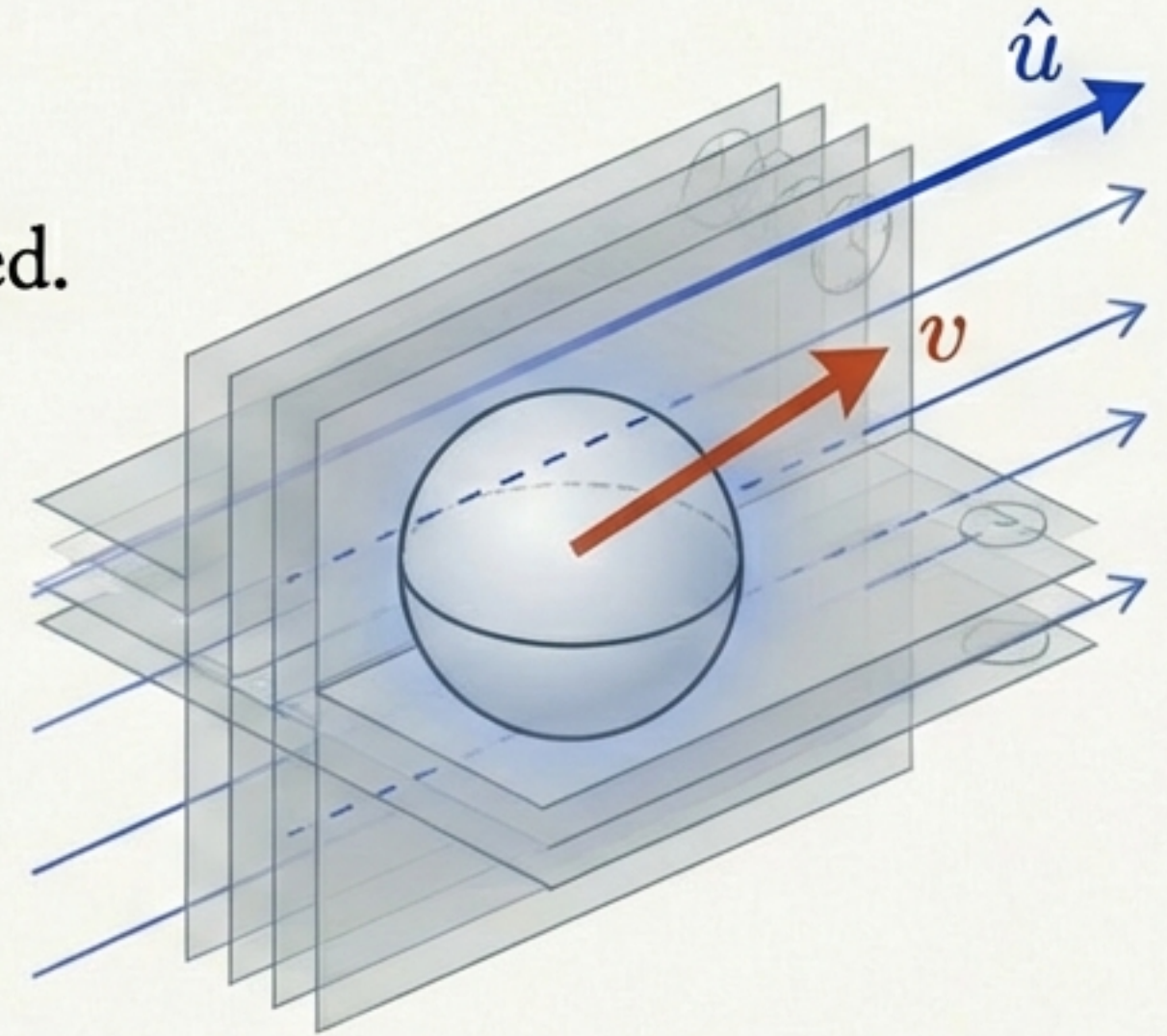
A system moving with spatial velocity v relative to the foliation samples a boosted slicing of the clock field. This results in an effective mass that varies with orientation and speed.

$$M_{eff}(v) = M_0 \left[1 + \alpha_1 \frac{v \cdot \hat{u}}{c} + \alpha_2 \frac{(v \cdot \hat{u})^2}{c^2} + \mathcal{O}\left(\frac{v^3}{c^3}\right) \right]$$

v : Spatial velocity relative to foliation

$\hat{u}$: Spatial direction of the clock field

α_1, α_2 : Coefficients quantifying the mass anisotropy

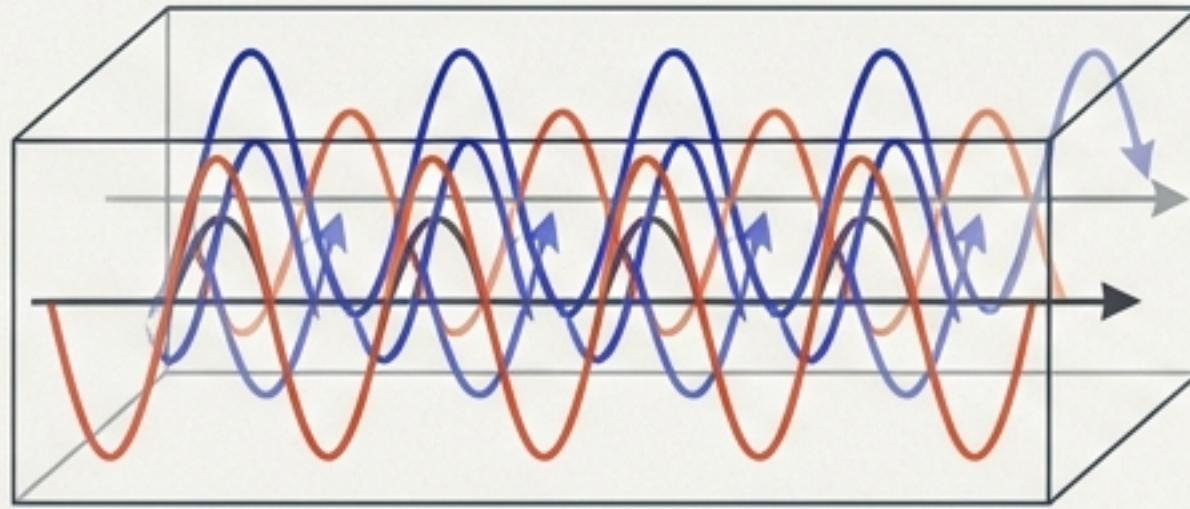


Key Distinction: These are mass corrections, not new forces. The inertia itself is changing.

Electromagnetic Non-Generation of Gravity

This effect is gravity-specific. Electromagnetic fields cannot directly modify gravitational mass unless they generate a persistent foliation pressure.

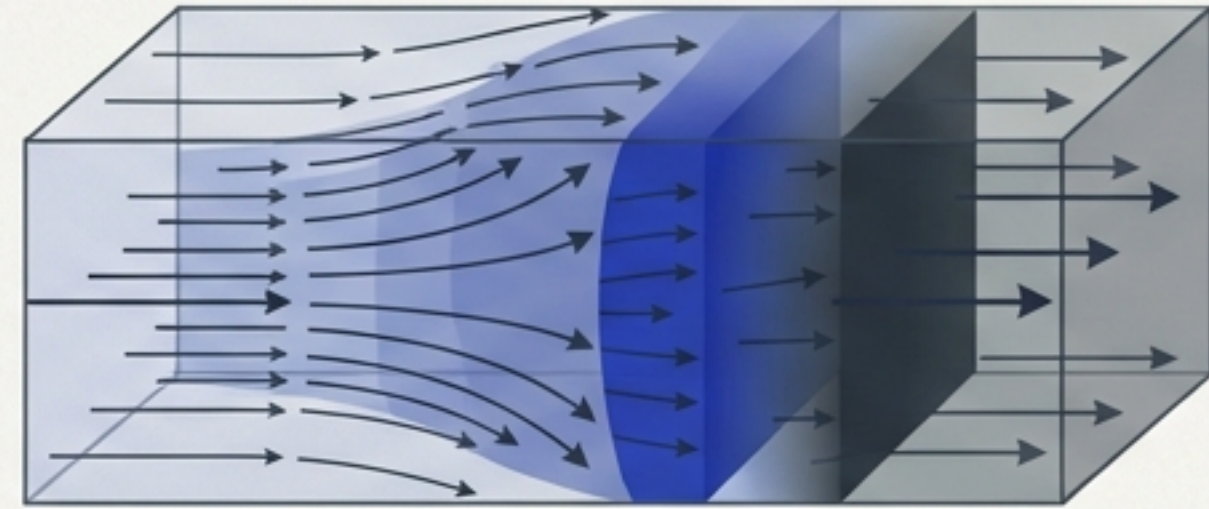
Radiative EM Energy (Lasers/RF)



$$\Delta(\nabla_{\mu}\chi) = 0$$

NO GRAVITY EFFECT

SST Mass Anomaly



$$\Delta(\nabla_{\mu}\chi) \neq 0$$

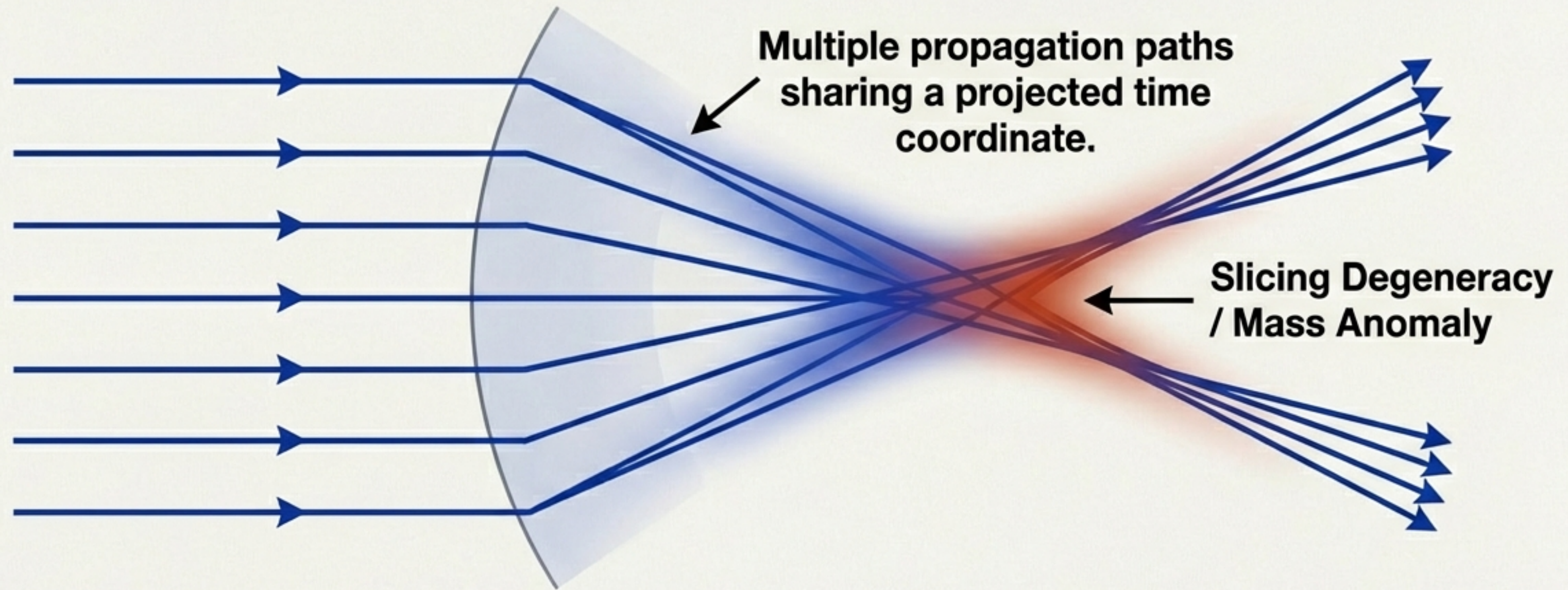
GRAVITATIONAL ANOMALY

Condition for gravitational effect: $\Delta(\nabla_{\mu}\chi) \neq 0$ after fields are removed.

Any anomaly detected in the upcoming experiment cannot be dismissed as a standard electromagnetic interaction; it must be a modification of the inertial mass functional itself.

Visualizing Foliation Degeneracies: The Phase Topology Analogy

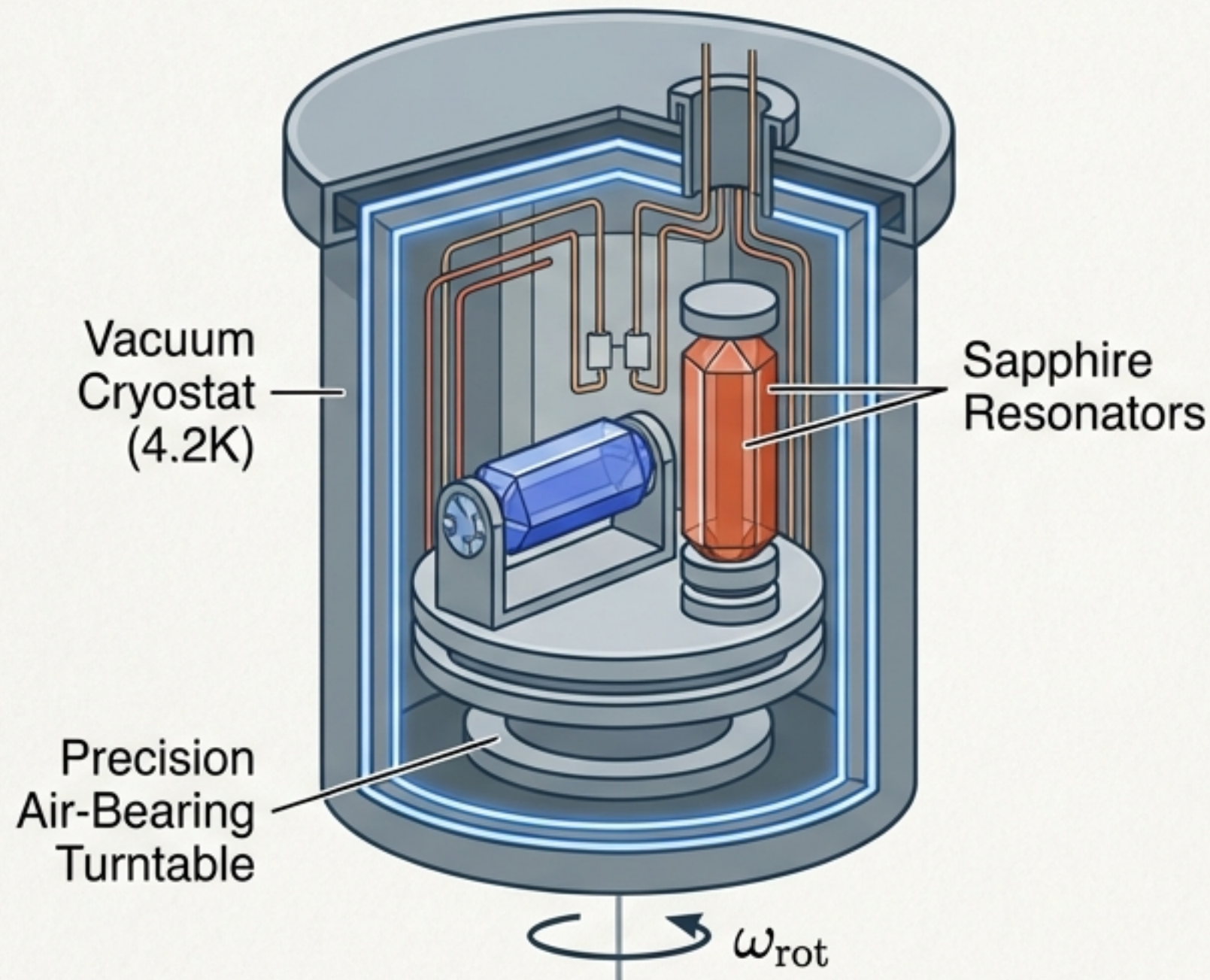
Just as wave-optical gravitational lensing creates interference patterns near caustics due to “phase geometry,” SST predicts mass anomalies due to “foliation slicing.”



Caustics represent degeneracies in the effective time-of-flight structure. In SST, motion creates similar “slicing degeneracies” in the clock field, altering the integrated energy (mass).

Experiment A: Rotating Cryogenic Sapphire Resonator

The Gravity-Sector Null Test



Physical Principle: If mass depends on velocity relative to the background foliation, the re-resonance frequencies of orthogonal crystals will diverge as they rotate.

The Signal Equation:

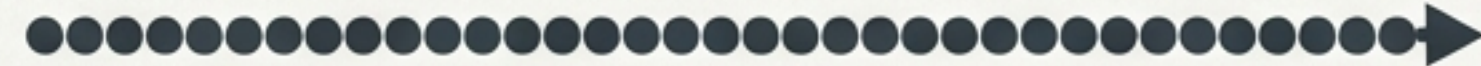
$$\frac{\Delta f}{f} = S_2 \cos(2\omega_{\text{rot}}t) + \sum S_{2,n} \cos(2\omega_{\text{rot}}t + n\omega_{\oplus}t)$$

Smoking Gun: Sidereal Sidebands. The signal is modulated by Earth's rotation ($\omega_{\oplus}$), uniquely identifying it as a preferred-frame effect.

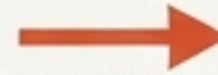
The Texture of Time: Relational and Countable

Time in SST is not a smooth external parameter. It is a “relational observable” derived from counting events in the medium.

Standard Time



Relational Time



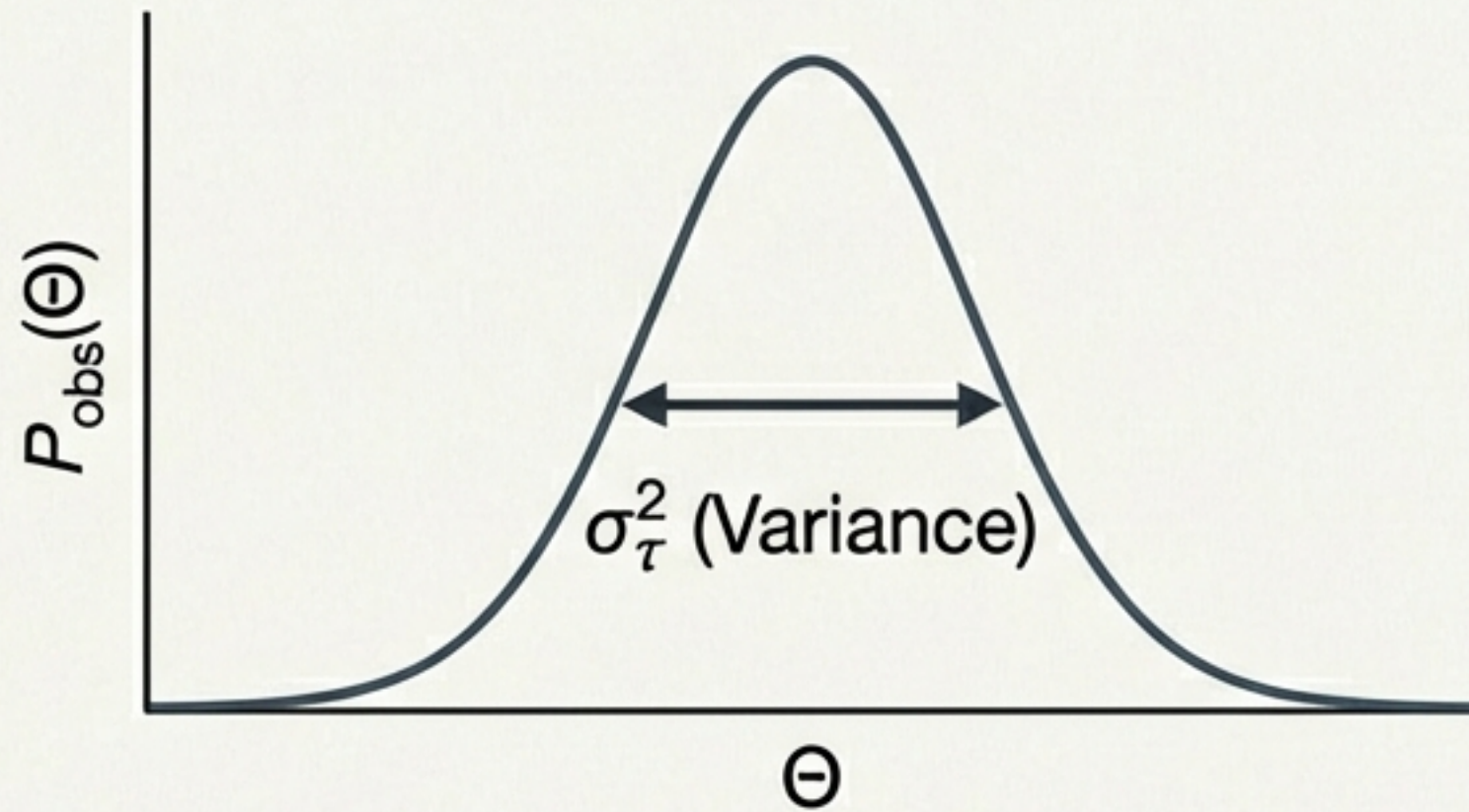
Deriving the Schrödinger Equation
(Appendix A):

$$i\hbar \frac{D\psi}{Dt} = -\frac{\hbar^2}{2m} \nabla^2_{\perp} \psi$$

Quantum Mechanics is not postulated; it is the effective wave dynamics of a system evolving along a preferred clock foliation u^{μ} .

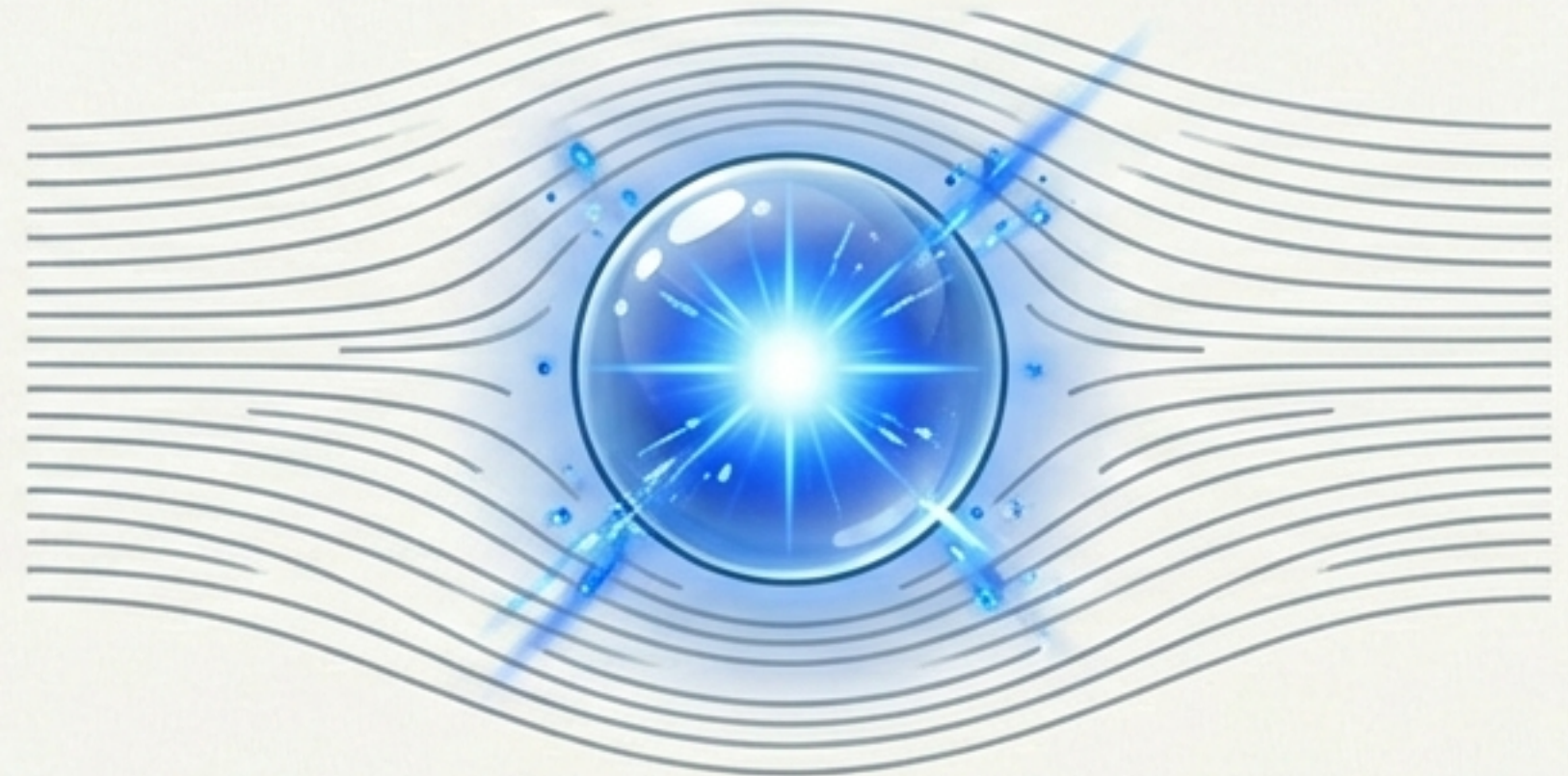
Intrinsic Temporal Stochasticity

Counting events is inherently noisy. Precise clock readouts must exhibit intrinsic fluctuations.



$$\sigma_{\tau}^2 = \sigma_{\tau}^2(\nabla \chi)$$

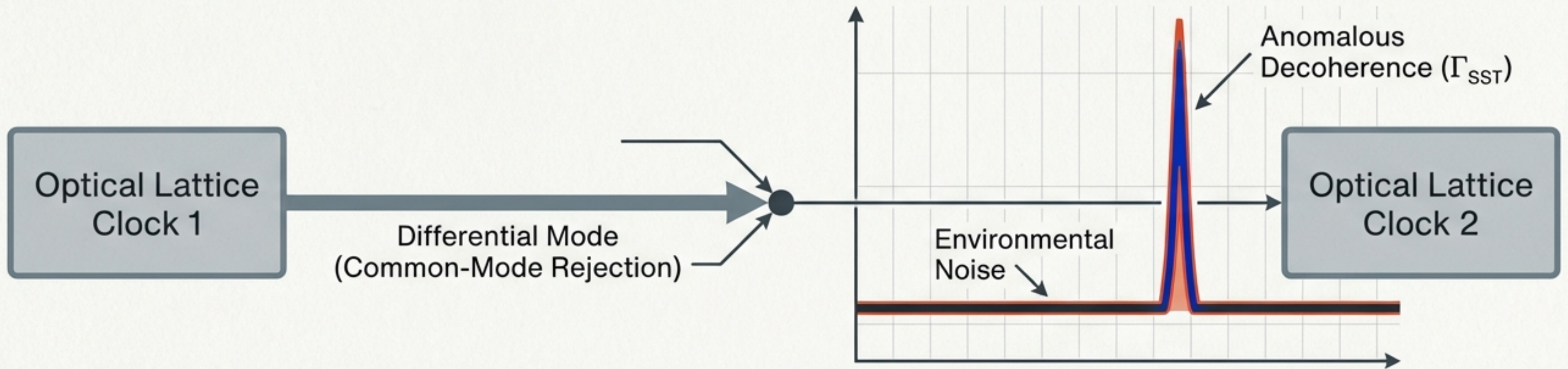
Sonoluminescence



Physical Analog: Single-bubble Sonoluminescence. Light emission timing varies based on internal phase organization, not just peak energy. This proves 'phase-dependent timing' is a real phenomenon in fluid dynamics.

Experiment B: The Clock-Noise Search

Operationalizing Intrinsic Temporal Stochasticity.

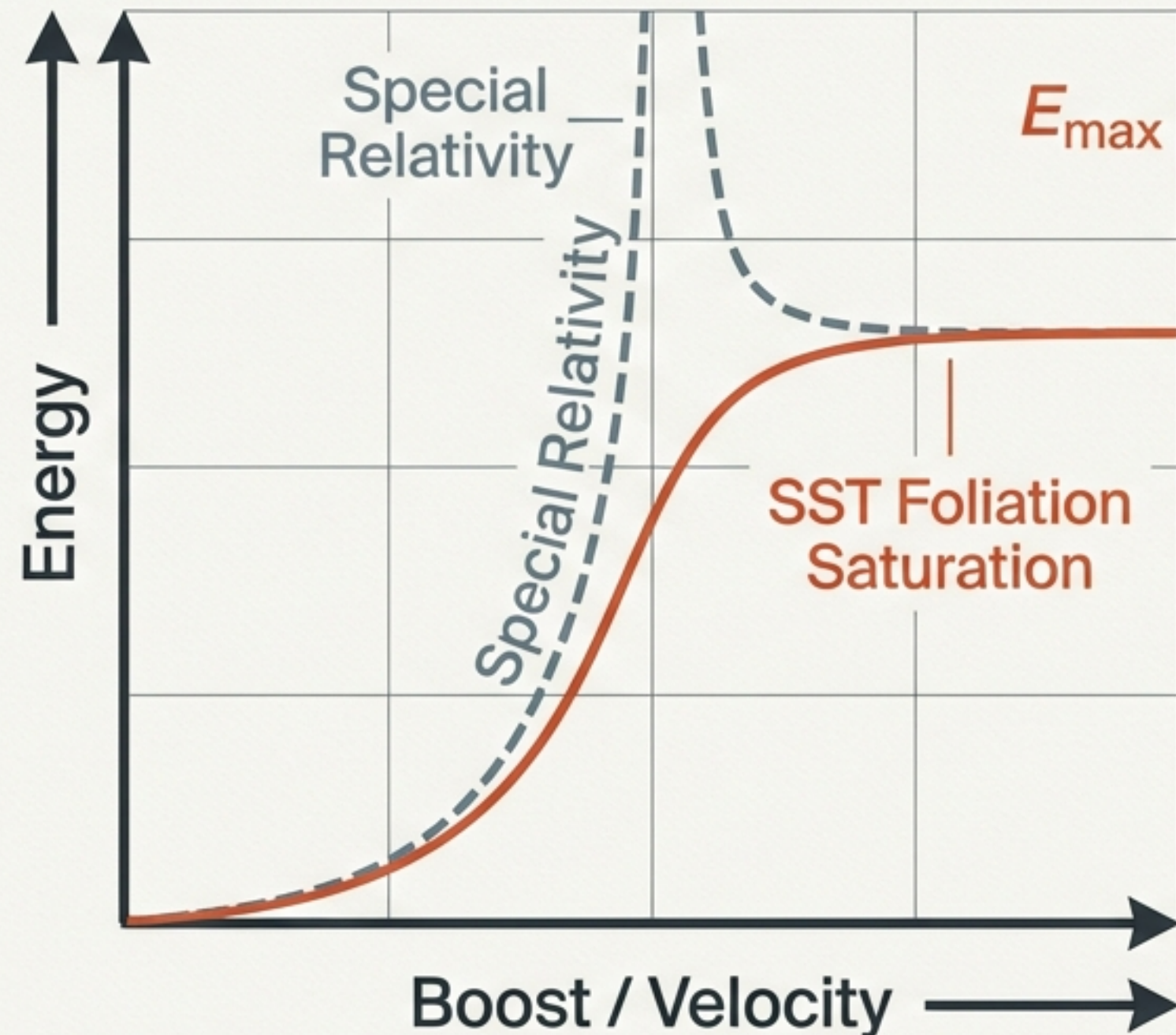


$$\Gamma_{\text{obs}} = \Gamma_{\text{env}} + \Gamma_{\text{SST}}$$

The SST Signal (Γ_{SST}) correlates with controlled modulation of the local clock-field gradient. This defines a specific Search Channel.

The Energy Ceiling: Foliation Saturation

Why Cosmic Rays Have a Maximum Energy



The Mäkelä Bound (Eq 45):

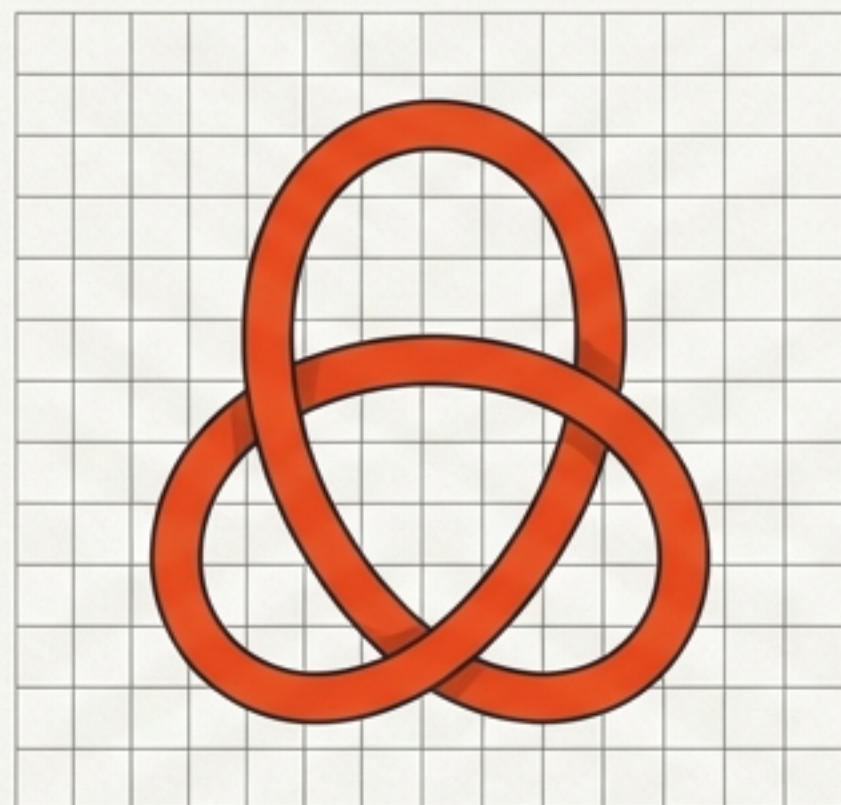
$$E_{\max} = Cc^2 \cosh \left(\frac{2\pi^2}{\ln 2} \right) \int_V \rho_E dV$$

Universal Factor $\gamma_{\max} \approx 1.17 \times 10^{12}$

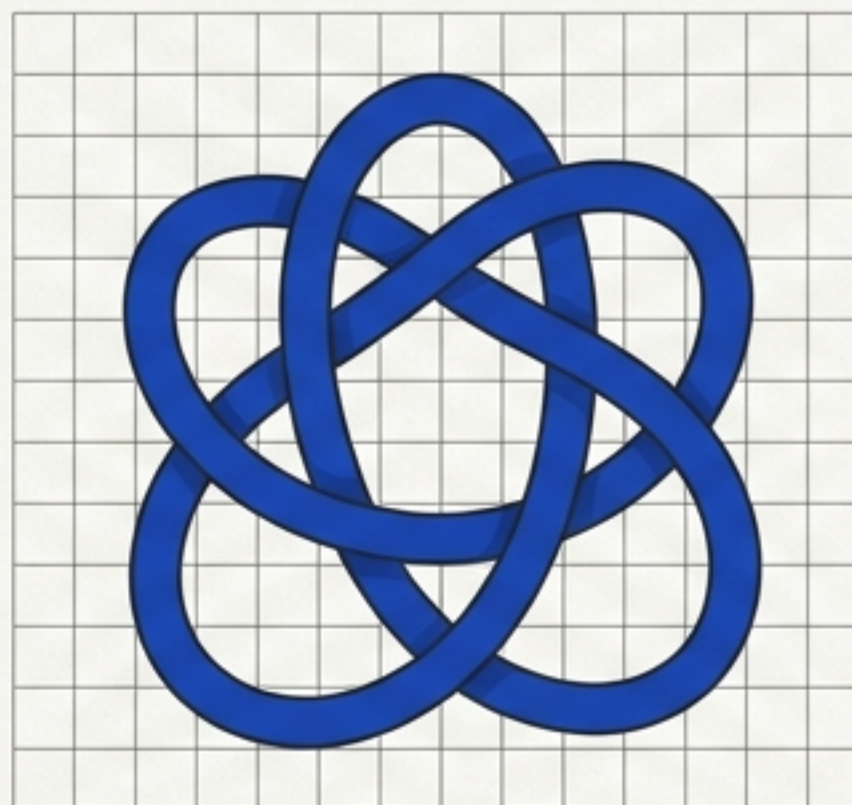
Energy saturation is a kinematic consequence of the compact clock foliation. Time effectively runs out of “boost”.

The Origin of Matter: Topology Over Symmetry

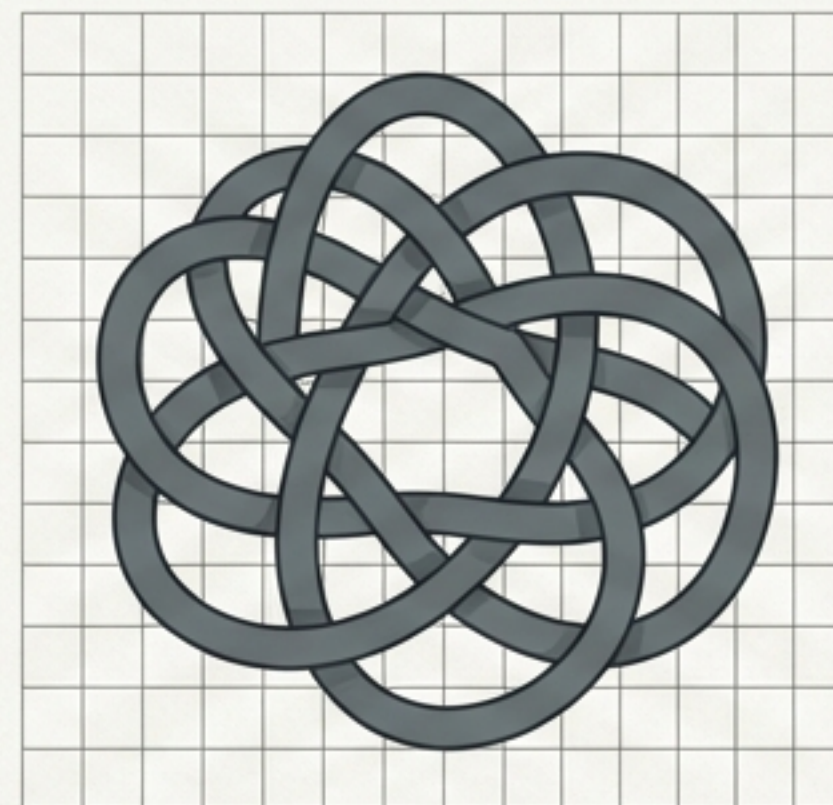
Replacing arbitrary Yukawa couplings with discrete topological order.



Top/Heavy Quark
(Simple Knot, High Energy Density)



Charm Quark



Up Quark
(Complex Knot, Diluted Energy)

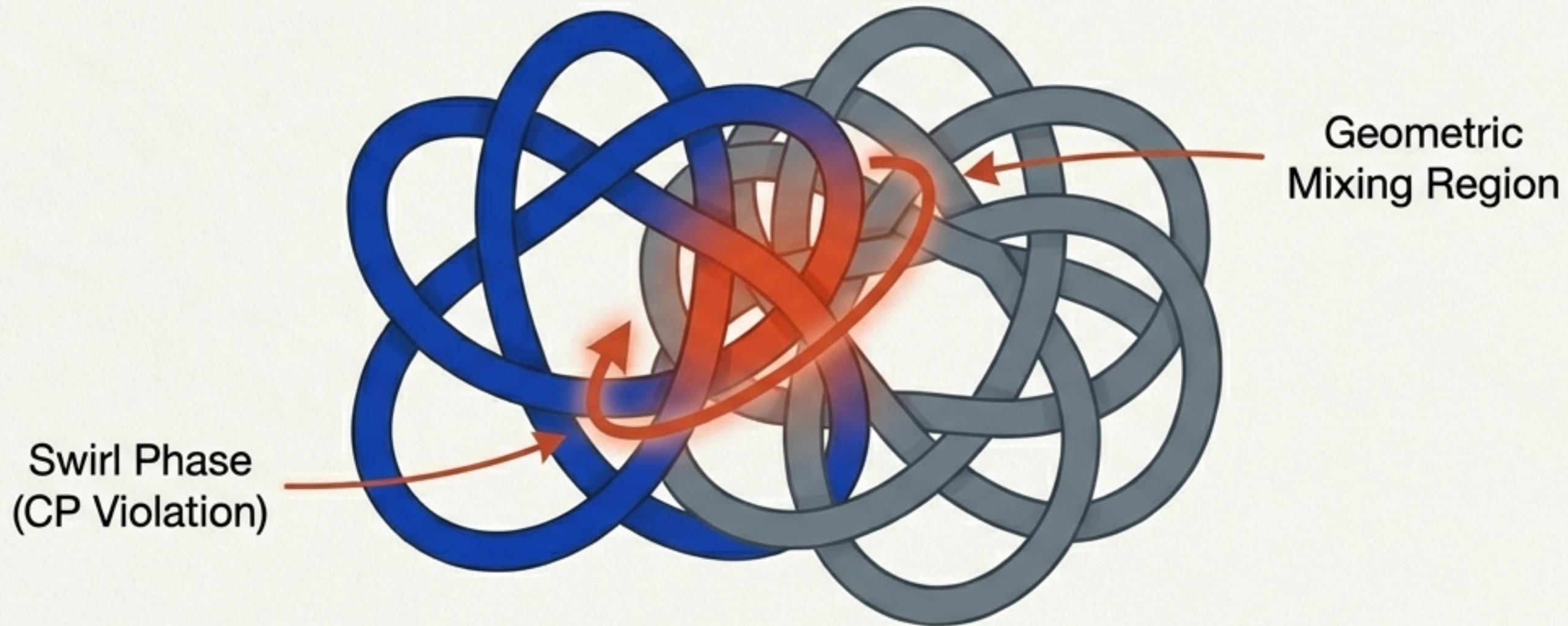
$$\frac{M^{(i)}}{M^{(j)}} \sim \epsilon_K^{n_i - n_j}$$

n_K : Topological complexity index.

Heavier fermions are simple knots. Lighter fermions are complex knots where energy is diluted over length.

Quark Mixing and Modular Symmetry

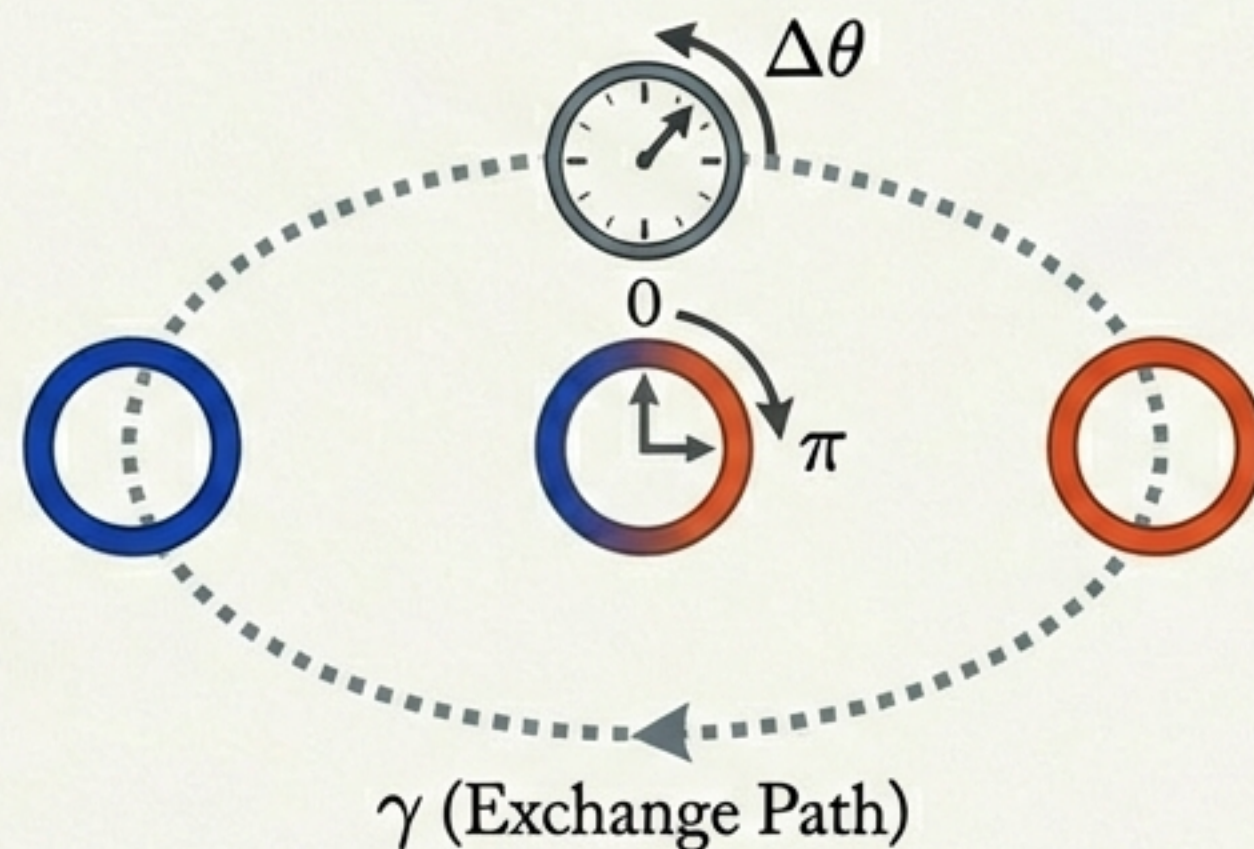
Flavor symmetries are the infrared shadow of Swirl-Clock geometry.



- Mixing: Arises from geometric overlap of swirl-string configurations.
- CP Violation: Emerges as a non-trivial 'swirl phase' (holonomy) between topological sectors.
- Modular Parameter τ : Represents the internal clock phase ($\theta_{\text{geom}} + i\theta_{\text{stab}}$).

Exchange Statistics as Swirl-Clock Holonomy

Why do particles behave as Bosons or Fermions?

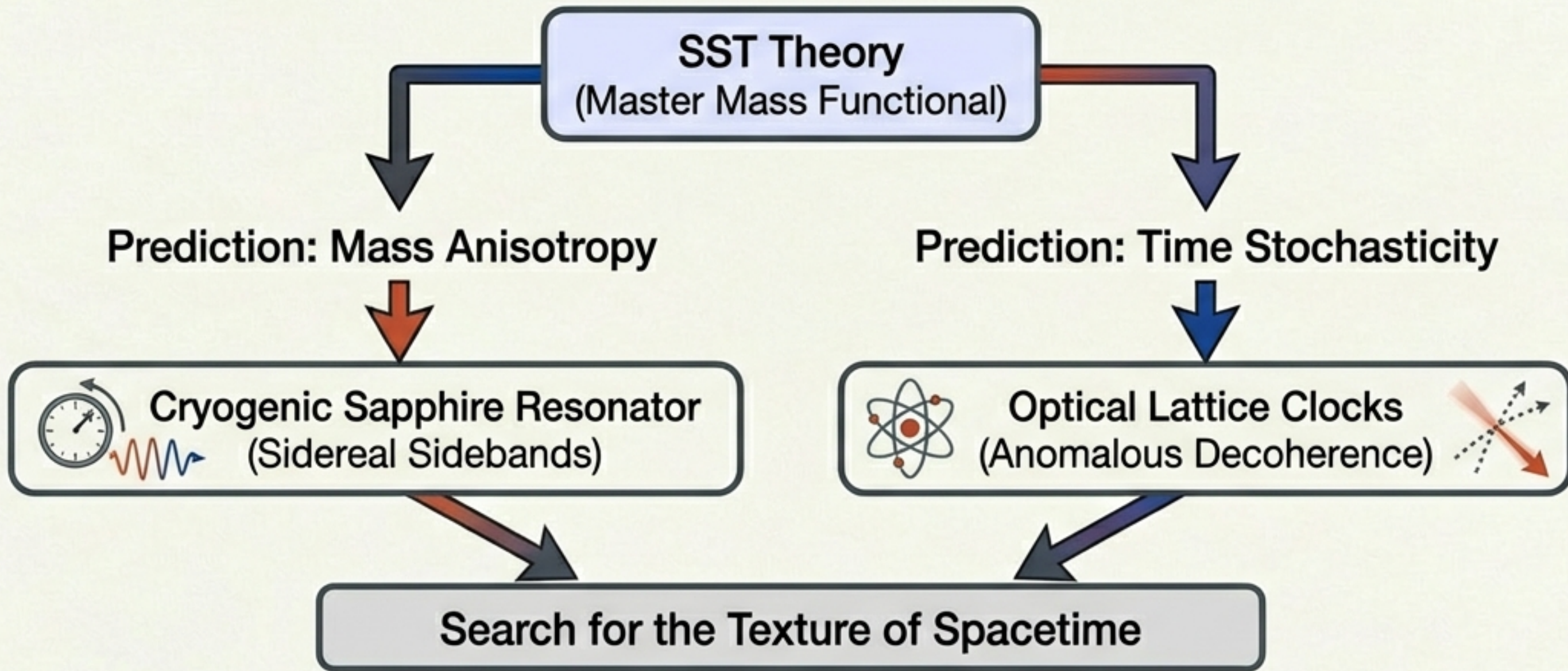


The exchange accumulates an internal clock phase $\Delta\theta = \oint dS_{\text{circle}}$.

- $\Delta\theta = 0$: Boson
- $\Delta\theta = \pi$: Fermion
- $\Delta\theta = \pi$: Fermion

Statistics are emergent. They depend on the topology of the exchange path and the memory encoded in the clock field.

Conclusion: A Minimal, Falsifiable Path Forward



- 1. SST unites Mass and Time under a single covariant action with a preferred foliation (u^μ).
- 2. Requires no new forces and respects GW170817.
- 3. Explains the origin of mass hierarchies and energy ceilings.

The proposed experiments define a search channel for the texture of spacetime itself.